

1. MC.09.02

Which of the following is part of the control process?

- ☐
- a. Monitoring of actual activity
- ☐
- b. Comparison of actual with planned activity
- ☐
- c. Investigating
- ☐
- d. Taking corrective action
- ☒
- e. All of these



Feedback

Check My Work

The control process consists of monitoring actual activity, comparing actual with planned outcomes, investigating unusual outcomes, and taking corrective action.

Solution
e

2. MC.09.03

Which of the following is *not* an advantage of budgeting?

- ☐
- a. It forces managers to plan.
- ☐
- b. It provides information for decision-making.
- ☒
- c. It guarantees an improvement in organizational efficiency.
- ☐
- d. It provides a standard for performance evaluation.
- ☐
- e. It improves communication and coordination.



Feedback

Check My Work

Advantages of budgeting are planning, information for decision makers, performance evaluation standards, and improved communication and coordination.

Solution
c

3. MC.09.04

The budget committee

- ☐
- a. reviews the budget.
- ☐
- b. resolves differences that arise as the budget is prepared.
- ☐
- c. approves the final budget.
- ☐
- d. is directed (typically) by the controller.
- ☒
- e. All of these.



Feedback

Check My Work

The budget committee reviews the budget, provides policy guidelines and budgetary goals, resolves differences that arise, approves final budget, and monitors actual performance.

Solution
e

4. MC.09.05

A moving, 12-month budget that is updated monthly is



- ☐ a. not used by manufacturing firms.
- ☐ b. a waste of time and effort.
- ☐ c. a master budget.
- ☒ d. a continuous budget.
- ☐ e. always used by firms that prepare a master budget.

Feedback

Check My Work

A continuous budget is a moving 12-month budget.

Solution
d

5. MC.09.06.ALGO (Algorithmic)

Which of the following is *not* part of the operating budget?



- ☐ a. The advertising budget
- ☒ b. The capital budget
- ☐ c. The production budget
- ☐ d. The direct labor budget
- ☐ e. The supplies budget

Feedback

Check My Work

The operating budget consists of the budgeted income statement and all supporting budgets and schedules.

Solution
b

6. MC.09.08.ALGO (Algorithmic)

The first step in preparing the sales budget is to



- ☒ a. prepare a sales forecast.
- ☐ b. review the company's strategic plan.
- ☐ c. review the production budget carefully.
- ☐ d. review staffing levels in the Sales department.
- ☐ e. estimate sales commissions.

Feedback

Check My Work

Review the preparation of the sales budget.

Solution
a

7. MC.09.10.ALGO (Algorithmic)

A company requires 90 pounds of plastic to meet the production needs of a small toy. It currently has 10 pounds of plastic inventory. The desired ending inventory of plastic is 30 pounds. How many pounds of plastic should be budgeted for purchasing during the coming period?

- ☐ a. 70 pounds

☐

b. 120 pounds

☒

c. 110 pounds

☐

d. 100 pounds

☐

e. None of these

Feedback

Check My Work

Correct

Post-Submission

$$90 + 30 - 10 = 110$$

Solution

c

8. MC.09.11

A company plans to sell 220 units. The selling price per unit is \$24. There are 50 units in beginning inventory, and the company would like to have 20 units in ending inventory. How many units should be produced for the coming period?

☐

a. 250

☐

b. 200

☐

c. 230

☐

d. 220

☒

e. None of these

Feedback

Check My Work

Correct

Post-Submission

$$220 + 20 - 50 = 190$$

Solution

e

9. MC.09.12

When a company amasses data during the preparation of the master budget, it is primarily doing so for which of the following purposes?

☐

a. Descriptive

☒

b. Predictive

☐

c. Prescriptive

☐

d. Diagnostic

Feedback

Check My Work

Correct.

Solution

b

10. MC.09.13

A company has the following collection pattern: month of sale, 40%; month following sale, 60%.

If credit sales for January and February are \$100,000 and \$200,000, respectively, the cash collections for February are?

☒

a. \$140,000.

- ☐ b. \$300,000.
- ☐ c. \$120,000.
- ☐ d. \$160,000.
- ☐ e. \$80,000.

Feedback

Check My Work
Correct
Post-Submission

$$(0.40 \times \$200,000) + (0.60 \times \$100,000) = \$140,000$$

Solution
a

11. MC.09.14

The percentage of accounts receivable that is uncollectible can be ignored for cash budgeting because

- ✓ ☒ a. no cash is received from an account that defaults.
- ☐ b. it is included in cash sales.
- ☐ c. it appears on the budgeted income statement.
- ☐ d. for most companies, it is not a material amount.
- ☐ e. None of these

Feedback

Check My Work

The amount a company deems for bad debt expense against receivables is not expected to be received in cash.

Solution
a

12. MC.09.15

Suppose that a factory increases training costs in order to reduce the waste of materials in production. That training cost is included in the

- ✓ ☒ d. the fixed factory overhead budget.
- ☐ a. sustainability budget.
- ☐ b. the variable selling expense budget.
- ☐ c. the direct materials purchases budget.
- ☐ e. the direct labor budget.

Feedback

Check My Work
Correct.
Solution
d

13. MC.09.16

Which of the following items is a possible example of myopic behavior?

- ☐ a. Failure to promote deserving employees
- ☐ b. Reducing expenditures on preventive maintenance
- ☐ c. Cutting back on new product development

-  ☐ d. Buying cheaper, lower-quality materials so that the company does not exceed the materials purchases budget
- ☐ e. All of these

Feedback

Check My Work

Myopic behavior occurs when actions affect short-term performance, but at the expense of the long-term budget.

Solution
e

14. MC.09.17

For performance reporting, it is best to compare actual costs with budgeted costs using

-  ☐ static budgets.
- ☐ master budgets.
- ☐ short-term budgets.
- ☒ flexible budgets.
- ☐ None of these

Feedback

Check My Work

A budget with financial results for a range of activity levels is best for performance reporting.

Solution
d

15. MC.09.18

To create a meaningful performance report, actual costs and expected costs should be compared

-  ☐ hourly.
- ☐ at the budgeted level of activity.
- ☐ weekly.
- ☒ at the actual level of activity.
- ☐ at the average level of activity.

Feedback

Check My Work

A meaningful performance report uses an after-the-fact budget. Review how after-the-fact budgets are prepared.

Solution
d

16. MC.09.19

To help assess performance, managers should use a

- ☐ static budget.
- ☐ before-the-fact flexible budget.
- ☐ master budget.
- ☐ continuous budget.



None of these

Feedback

Check My Work

A meaningful performance report compares actual costs to budgeted costs at actual levels of production.

Solution
e

17. MC.09.20

A firm comparing the actual variable costs of producing 10,000 units with the total variable costs of a static budget based on 9,000 units would probably notice

- ☐ large favorable variances.
- ☐ no variances.
-  ☒ large unfavorable variances.
- ☐ small unfavorable variances.
- ☐ small favorable variances.

Feedback

Check My Work

Decide if the output causes a variance assuming a static budget. If so, is the variance favorable or unfavorable, meaning more or less costly than budget? A variance of 10% or more is considered large.

Solution
c

18. MC.09.61

Alpha Omega Enterprises budgeted the following sales in units:

January 40,000
February 30,000
March 50,000

Alpha Omega's policy is to have 30% of the following month's sales in inventory. On January 1, inventory equaled 10,000 units. February production in units is:

- ☐ a. 45,000.
- ☐ b. 36,000.
-  ☒ c. 53,000.
- ☐ d. 20,000.

Feedback

Check My Work
Correct
Post-Submission

Units to Be Produced = Expected Unit Sales + Units in Desired Ending Inventory (EI) – Units in Beginning Inventory (BI)

	<u>January</u>	<u>February</u>
Expected unit sales	40,000	50,000
+ Units in desired ending inventory (EI) (Following month's sales in units × 30%)	<u>15,000*</u>	<u>18,000**</u>
Units needed	55,000	68,000
– Units in beginning inventory (BI)	<u>10,000</u>	<u>15,000</u>
Units to be produced	<u>45,000</u>	<u>53,000</u>

*January: Units in Desired Ending Inventory (EI) = 50,000 units × 30% = 15,000 units

**February: Units in Desired Ending Inventory (EI) = 60,000 units × 30% = 18,000 units

19. MC.09.63

A company expects the following sales for the coming year:

	<u>1st Quarter</u>	<u>2nd Quarter</u>	<u>3rd Quarter</u>	<u>4th Quarter</u>
Units	40,000	60,000	80,000	100,000
Average selling price	\$6	\$6	\$6	\$8

Budgeted sales revenue for the year is:

-  ☐ a. \$1,050,000.
- ☒ b. \$1,880,000.
- ☐ c. \$1,260,000.
- ☐ d. \$1,650,000.

Feedback

Check My Work
Correct
Post-Submission

Budgeted Sales Revenue = [(Expected Sales in Units for 1st Quarter + Expected Sales in Units for 2nd Quarter + Expected Sales in Units for 3rd Quarter) × Average Selling Price] + (Expected Sales in Units for 4th Quarter × Average Selling Price) = [(40,000 units + 60,000 units + 80,000 units) × \$6] + (100,000 units × \$8) = \$1,880,000

20. MC.09.67

Unique Company provided the following budgeted data for July:

Direct materials	\$70,000
Direct labor	\$45,000
Overhead	\$110,000
Beginning finished goods	\$20,000
Ending finished goods	\$35,000
Production in units	15,000

What is the budgeted cost of goods sold?

- ☐ a. \$179,000
- ☐ b. \$184,000

- ☒ c. \$165,000
- ☐ d. \$210,000

Feedback

Check My Work
Correct
Post-Submission

Budgeted Manufacturing Costs = Direct Materials Used + Direct Labor Used + Overhead = \$70,000 + \$45,000 + \$110,000 = \$225,000
Budgeted Cost of Goods Sold = Budgeted Manufacturing Costs + Beginning Finished Goods – Ending Finished Goods = \$225,000 + \$20,000 – \$35,000 = \$210,000

21. MC.09.80

A company plans to sell 37,000 units of Product No. 99 in November, and each of these units requires five square feet of raw material. Pertinent data follow:

Product No. 99 Raw Materials

Actual November 1 inventory	7,000	10,800 square feet
Estimated November 30 inventory	5,800	? square feet

If the company purchases 205,000 square feet of raw materials during the month, the estimated raw materials inventory on November 30 would be:

- ☒ a. 36,800 square feet.
- ☐ b. 53,000 square feet.
- ☐ c. 42,800 square feet.
- ☐ d. 41,000 square feet.

Feedback

Check My Work
Correct
Post-Submission

$37,000 - 7,000 + 5,800 = 35,800$ finished; $35,800 \times 5 = 179,000$ units of raw materials; $179,000 - 10,800 = 168,200$ units of raw materials; $205,000 - 168,200 = 36,800$ sq. ft. at November 30.

22. MC.09.85

A company produces two products: SR200 and TX500. Budgeted sales for four months are as follows:

SR200 TX500

May	8,000	20,000
June	13,000	32,000
July	11,000	39,000
August	18,000	46,000

The company's ending inventory policy is that SR200 should have 15% of next month's sales in ending inventory and TX500 should have 20% of next month's sales in ending inventory. On May 1, there were 1,500 units of SR200 and 8,000 units of TX500.

TX500 requires six units of Component A. (SR200 does not use Component A.) There were 25,000 units of Component A in inventory on May 1. Kenner wants to have 20% of the following month's production needs in inventory for Component A. What is the desired ending inventory of

Component A for May?



- ☐ a. 133,400
- ☐ b. 25,000
- ☒ c. 40,800
- ☐ d. 125,480

Feedback

Check My Work
Correct
Post-Submission

	<u>May</u>	<u>June</u>
Production	18,400	33,400
× Units of Component A	<u> × 6 </u>	<u> × 6 </u>
Component A needed for production	110,400	<u>200,400</u>
Desired ending inventory*	<u>40,080</u>	
Units needed	158,400	
Less: Beginning inventory	<u>(25,000)</u>	
Production	<u>133,400</u>	

*(33,400 × 6) × 20% = 40,080

23. MC.09.89

A company plans to sell 142,000 units in November and 178,000 units in December. The company's policy is that 10% of the following month's sales must be in ending inventory. On November 1, there were 14,000 units in inventory.

It takes 30 minutes of direct labor time to make one unit. Direct labor wages average \$18 per hour. Variable overhead is applied at the rate of \$5 per direct labor hour. Fixed overhead is budgeted at \$57,000 per month. What is the direct labor cost budgeted for November?



- ☒ a. \$1,312,200
- ☐ b. \$2,152,000
- ☐ c. \$950,600
- ☐ d. \$1,181,500

Feedback

Check My Work
Correct
Post-Submission

November Production = 142,000 + (0.10 × 178,000) – 14,000 = 145,800 units
Direct Labor Hours = 0.50 hour per unit × 145,800 = 72,900 hours
Direct Labor Cost = \$18 × 72,900 = \$1,312,200

24. MC.09.93

A company provided the following information for last month:

Production in units	5,000
Direct materials cost	\$8,000
Direct labor cost	\$12,000
Overhead cost	\$9,600

Sales commission per unit sold	\$5
Price per unit sold	\$30
Fixed selling and administrative expense	\$8,000

There were no beginning and ending inventories. What is the gross margin for the company last month?

- ☐ a. \$80,400
- ☐ b. \$105,600
- ☐ c. \$60,500
- ☒ d. \$120,400



Feedback

Check My Work
Correct
Post-Submission

Sales (\$30 × 5,000)	\$87,000
Cost of goods sold (\$8,000 + \$12,000 + \$9,600)	<u>29,600</u>
Gross margin	<u>\$120,400</u>

25. MC.09.97

A company produces speaker systems for cars. Estimated sales (in units) in January are 40,000; in February are 37,000; and in March are 34,000. Each unit is priced at \$56. The company wants to have 30% of the following month's sales in ending inventory. That requirement was met on January 1.

Each speaker system requires 3 boxes and 15 yards of wire. Boxes cost \$5 each, and wire is \$0.62 per yard. The company wants to have 20% of the following month's production needs in ending raw materials inventory. On January 1, the company had 24,000 boxes and 100,000 yards of wire in inventory. How many boxes does the company expect to purchase in January?

- ☐ a. 148,500
- ☐ b. 116,260
- ☐ c. 196,550
- ☒ d. 108,960



Feedback

Check My Work
Correct
Post-Submission

	<u>January</u>	<u>February</u>
Production	37,100	36,100
× Raw materials per unit	<u>× 3</u>	<u>× 3</u>
Raw materials for production	111,300	<u>108,300</u>
Desired ending inventory	<u>21,600</u>	
Units needed	132,960	
Less: Beginning inventory	<u>(24,000)</u>	
Purchases in units	<u>108,960</u>	

26. MC.09.102

A sporting goods store sells bicycles throughout the southeastern United States. The following data were taken from the most recent quarterly sales forecast:

	End-of-Month	
	Expected Sales	Target Inventory
July	1,990 units	400 units
August	2,150 units	490 units
September	2,070 units	460 units

On the basis of the information presented, how many bicycles should the company purchase in August?

- 
- ☐

a. 2,630
- ☐

b. 2,040
- ☐

c. 2,330
- ☒

d. 2,240

Feedback

Check My Work

Correct

Post-Submission

$$(2,150 + 490) - 400 = 2,240 \text{ units purchased in August}$$

27. MC.09.112

A company makes all sales on account, subject to the following collection pattern: 30% are collected in the month of sale; 60% are collected in the first month after sale; and 10% are collected in the second month after sale. If sales for October, November, and December were \$78,000, \$68,000, and \$48,000, respectively, what was the budgeted receivables balance on December 31?

- 
- ☒

a. \$40,400
- ☐

b. \$37,400
- ☐

c. \$56,000
- ☐

d. \$42,600

Feedback

Check My Work

Correct

Post-Submission

$$\begin{aligned} \text{Receivables} - \text{Collected} &= (\$78,000 + \$68,000 + \$48,000) - [(\$78,000 \times 100\%) + (\$68,000 \times 90\%) + (\$48,000 \times 30\%)] = \\ & \$194,000 - \$153,600 = \$40,400 \end{aligned}$$

28. MC.09.128

A company makes all its sales on account. The company's accounts receivable payment experience is as follows:

Percent paid in the month of sale	30%
Percent paid in the month after the sale	65%
Percent paid in the second month after the sale	3%

The company provided the following information on sales:

September	\$110,000
-----------	-----------

October	120,000
November	230,000
December (expected)	260,000

What are the expected cash receipts in the month of November?

- 
- ☐ a. \$190,000
- ☒ b. \$150,300
- ☐ c. \$160,600
- ☐ d. \$40,200

Feedback

Check My Work
Correct
Post-Submission

Expected Cash Receipts in November = (Sales Made in November × Percent Paid in Month of Sale) + (Sales Made in October × Percent Paid in Month after Sale) + (Sales Made in September × Percent Paid in Second Month after Sale) = (\$230,000 × 0.30) + (\$120,000 × 0.65) + (\$110,000 × 0.03) = \$69,000 + \$78,000 + \$3,300 = \$150,300

29. MC.09.144
Jason Inc. produces leather purses. Jason has developed a static budget for the first quarter based on 20,000 direct labor hours. During the quarter, the actual activity was 22,000 direct labor hours. Data for the first quarter are summarized as follows:

	<u>Static Budget</u> <u>(20,000 hours)</u>	<u>Actual Costs</u> <u>(22,000 hours)</u>
Direct materials cost	\$ 80,000	\$ 87,000
Direct labor cost	160,000	174,000
Building rental	<u>48,000</u>	<u>50,000</u>
Total	<u>\$288,000</u>	<u>\$311,000</u>

What is the flexible budget variance for the first quarter?

- 
- ☐ a. \$23,000 F
- ☐ b. \$23,000 U
- ☐ c. \$1,000 U
- ☒ d. \$1,000 F

Feedback

Check My Work
Correct
Post-Submission

	<u>Flexible Budget</u> <u>(22,000 hours)</u>	<u>Actual Costs</u> <u>(22,000 hours)</u>	<u>Variance</u>
Direct materials	\$ 88,000	\$ 87,000	\$1,000 F
Direct labor	176,000	174,000	2,000 F
Building rental	<u>48,000</u>	<u>50,000</u>	<u>2,000</u> U
Total	<u>\$312,000</u>	<u>\$311,000</u>	<u>\$1,000</u> F

The flexible budget here is created by creating standard rates from the static budget. \$80,000 or direct labor divided by 20,000 labor hours yields \$4 per labor hour.
This standard rate is multiplied by the actual labor hours to get the standard labor hours for the actual activity level.

30. MC.09.171

A company budgeted units to be sold for each quarter of the coming year:

- Quarter 1 1,200
- Quarter 2 1,500
- Quarter 3 1,800
- Quarter 4 2,000

The company has a policy that requires 20% of the next quarter's sales in ending inventory. Beginning inventory for the first quarter was 150. Sales for the first quarter of the year after the budget are estimated at 2,200 units. What is the desired ending inventory for the third quarter?

- ✓
- ☒ a. 440 units
- ☐ b. 360 units
- ☐ c. 600 units
- ☐ d. 280 units

Feedback

Check My Work
Correct
Post-Submission

Ending Inventory, Quarter 4 = 20% of Quarter after the Year Budget Sales
 $0.20 \times 2,200 = 440$ units

31. PR.09.221.ALGO (Algorithmic)

Bottleneck Company has put together the following data in order to complete its operating budget for the second quarter in the current year:

	April	May	June	July
Sales (units)	73,000	69,700	66,000	67,900

Additional information:
Company policy requires that 68% of next month's sales (in units) be in ending inventory. This policy was met in March.
It takes 3 hours of direct labor to produce one unit.
The average wage cost is \$14.
Variable overhead rate is \$7 per direct labor hour, and fixed overhead is \$10,500 per month.

Required:

A. Prepare a production budget for April, May, June and the quarter in total.

Bottleneck Company Production Budget For the Quarter Ended June 30				
	April	May	June	Total
Sales in units	✓ 73000	✓ 69700	✓ 66000	✓ 208700
Desired ending inventory	✓ 47396	✓ 44880	✓ 46172	✓ 46172

Total needs	✓ 120396	✓ 114580	✓ 112172	✓ 254872
Less: Beginning inventory	✓ 49640	✓ 47396	✓ 44880	✓ 49640
Units to be produced	✓ 70756	✓ 67184	✓ 67292	✓ 205232

Feedback

Check My Work
Correct
Post-Submission

A.

Desired ending inventory:

April ($69,700 \times 0.68$) = 47,396

May ($66,000 \times 0.68$) = 44,880

June ($67,900 \times 0.68$) = 46,172

Less: Beginning inventory:

April ($73,000 \times 0.68$) = 49,640

May ($69,700 \times 0.68$) = 47,396

June ($66,000 \times 0.68$) = 44,880

B. Prepare a direct labor budget for April, May, June and the quarter in total.

Bottleneck Company
Direct Labor Budget
For the Quarter Ended June 30

	April	May	June	Total
Units to be produced	✓ 70756	✓ 67184	✓ 67292	✓ 205232
Direct labor hours per unit	✓ 3	✓ 3	✓ 3	✓ 3
Total hours needed	✓ 212268	✓ 201552	✓ 201876	✓ 615696
Average wage rate per hour	\$ ✓ 14	\$ ✓ 14	\$ ✓ 14	\$ ✓ 14
Total direct labor cost	\$ ✓ 2971752	\$ ✓ 2821728	\$ ✓ 2826264	\$ ✓ 8619744

Feedback

Check My Work
Correct

C. Prepare an overhead budget for April, May, June and the quarter in total.

Bottleneck Company
Overhead Budget
For the Quarter Ended June 30

	<u>April</u>	<u>May</u>	<u>June</u>	<u>Total</u>
Budgeted direct labor hours	212268 ✓	201552 ✓	201876 ✓	615696 ✓
Variable overhead rate	\$ 7 ✓	\$ 7 ✓	\$ 7 ✓	\$ 7 ✓
Budgeted variable overhead	\$ 1485876 ✓	\$ 1410864 ✓	\$ 1413132 ✓	\$ 4309872 ✓
Budgeted fixed overhead	10500 ✓	10500 ✓	10500 ✓	31500 ✓
Total overhead	\$ 1496376 ✓	\$ 1421364 ✓	\$ 1423632 ✓	\$ 4341372 ✓

Feedback

Check My Work
Correct

Feedback

Check My Work
Correct
Solution

A. Prepare a production budget for April, May, June and the quarter in total.

Bottleneck Company
Production Budget
For the Quarter Ended June 30

	<u>April</u>	<u>May</u>	<u>June</u>	<u>Total</u>
1 Sales in units	73000	69700	66000	208700
Desired ending inventory	47396	44880	46172	46172
Total needs	120396	114580	112172	254872
Less: Beginning inventory	49640	47396	44880	49640
Units to be produced	70756	67184	67292	205232

B. Prepare a direct labor budget for April, May, June and the quarter in total.

Bottleneck Company
Direct Labor Budget
For the Quarter Ended June 30

	<u>April</u>	<u>May</u>	<u>June</u>	<u>Total</u>
2 Units to be produced	70756	67184	67292	205232
Direct labor hours per unit	3	3	3	3
Total hours needed	212268	201552	201876	615696
Average wage rate per hour	\$ 14	\$ 14	\$ 14	\$ 14
Total direct labor cost	\$ 2971752	\$ 2821728	\$ 2826264	\$ 8619744

3

C. Prepare an overhead budget for April, May, June and the quarter in total.

Bottleneck Company
Overhead Budget

For the Quarter Ended June 30

	<u>April</u>	<u>May</u>	<u>June</u>	<u>Total</u>
Budgeted direct labor hours	212268	201552	201876	615696
Variable overhead rate	\$ 7	\$ 7	\$ 7	\$ 7
Budgeted variable overhead	\$ 1485876	\$ 1410864	\$ 1413132	\$ 4309872
Budgeted fixed overhead	10500	10500	10500	31500
Total overhead	\$ 1496376	\$ 1421364	\$ 1423632	\$ 4341372

32. PR.09.222.ALGO (Algorithmic)

Jones Corporation has the following budgeted sales for the selected four-month period:

<u>Month</u>	<u>Unit Sales</u>
July	19,300
August	35,600
September	26,000
October	31,800

Sales price per unit is \$160.

Plans are to have an inventory of finished product equal to 24% of the unit sales for the next month. There were 2,800 units in beginning inventory on July 1.

2 pounds of materials are required for each unit produced. Each pound of material costs \$15. Inventory levels for materials equal 26% of the needs for the next month.

Desired ending inventory for September is 25,600 pounds of material. Beginning inventory for July was 20,100 pounds of material.

Each unit requires 1.6 hours of direct labor, and the average wage rate is \$6 per hour.

Variable overhead rate is \$1.00 per direct labor hour. There is also fixed overhead of \$19,500 per month.

The company pays a 4% commission on sales.

The Company has fixed selling and administrative expenses as follows:

Rent	\$3,600/month
Utilities	\$1,000/month
Advertising	\$200/month
Office salaries	\$36,500/month

Required:

Unless otherwise noted, round calculations to the nearest whole dollar. And use rounded amounts in subsequent calculations. Enter all final answers to the nearest whole dollar unless stated to do otherwise.

A. Prepare a sales budget for July, August, and September, and in total for the quarter.

Jones Corporation
Sales Budget

<u>July</u>	<u>August</u>	<u>September</u>	<u>Total</u>
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Units to be sold	19300	35600	26000	80900
	\$	\$	\$	\$
Sales price	160	160	160	160
	\$	\$	\$	\$
Total sales dollars	3088000	5696000	4160000	12944000

Feedback

Check My Work
Correct

B. Prepare a production budget for July, August, and September, and in total for the quarter.

Jones Corporation
Production Budget

	July	August	September	Total
Sales	19300	35600	26000	80900
Add: Desired ending inventory	8544	6240	7632	7632
Total needs	27844	41840	33632	88532
Less: Beginning inventory	4632	8544	6240	4632
Units to be produced	23212	33296	27392	83900

Feedback

Check My Work
Correct

C. Prepare a direct materials purchases budget in pounds and dollars for July, August, and September, and in total for the quarter.

Jones Corporation
Direct Materials Purchases Budget

	July	August	September	Total
Units to be produced	23212	33296	27392	83900
x lbs. per unit	2	2	2	2
Pounds needed	46424	66592	54784	167800
Desired ending inventory	17314	14244	25600	25600

Total needs	63738	80836	80384	193400
	✓	✓	✓	✓
Less: Beginning inventory	20100	17314	14244	20100
	✓	✓	✓	✓
Purchases needed	43638	63522	66140	173300
	✓	✓	✓	✓
x \$15/lb.	15	15	15	15
	✓	✓	✓	✓
Total purchase cost	654570	952830	992100	2599500
	✓	✓	✓	✓

Feedback

Check My Work
Correct

D. Prepare a direct labor budget in hours and total cost for July, August, and September, and in total for the quarter.

Jones Corporation
Direct Labor Budget

	July	August	September	Total
	✓	✓	✓	✓
Units to be produced	23212	33296	27392	83900
	✓	✓	✓	✓
x 1.6 of direct labor hours per unit	1.60	1.60	1.60	1.60
	✓	✓	✓	✓
Direct labor hours needed	37139	53274	43827	134240
	✓	✓	✓	✓
x \$6 per hour	6	6	6	6
	✓	✓	✓	✓
Total direct labor cost	222834	319644	262962	805440
	✓	✓	✓	✓

Feedback

Check My Work
Correct

E. Prepare an overhead budget for July, August, and September, and in total for the quarter.

Jones Corporation
Overhead Budget

	July	August	September	Total
	✓	✓	✓	✓
Labor hours	37139	53274	43827	134240
	✓	✓	✓	✓
x Variable rate per hour	1.00	1.00	1.00	1.00
	✓	✓	✓	✓

	\$	\$	\$	\$
Total variable OH	37139	53274	43827	134240
Fixed overhead	19500	19500	19500	58500
Total overhead	56639	72774	63327	192740

Feedback

Check My Work
Correct

F. Prepare a selling and administrative expenses budget for July, August, and September, and in total for the quarter.

Jones Corporation Selling and Administrative Expenses Budget				
	July	August	September	Total
	\$	\$	\$	\$
Variable selling expense	123520	227840	166400	517760
Fixed expenses:				
Rent	3600	3600	3600	10800
Utilities	1000	1000	1000	3000
Advertising	200	200	200	600
Office salaries	36500	36500	36500	109500
Total	164820	269140	207700	641660

Feedback

Check My Work
Correct

G. Prepare an ending finished goods inventory budget for the quarter. **[Hint: You have already calculated the desired ending finished goods inventory quantity. Assume a stable per-unit rate. (Round the per-unit fixed factory overhead rate to two decimal places.)]**

Jones Corporation Ending Finished Goods Inventory Budget	
	\$
Desired ending inventory	320544
	\$
Direct materials	30

	\$
Direct labor	9.60
Overhead:	
	\$
Variable overhead	1.60
	\$
Fixed overhead	0.70
	\$
Unit cost	42.00

Feedback

Check My Work
Correct
Post-Submission

G.

Desired ending inventory $(7,632 \times \$42) = \$320,544$
 Direct materials $(\$15 \times 2) = \30.00
 Direct labor $(1.60 \times \$6) = \9.6
 Variable overhead $(1.60 \times \$1) = \1.60
 Fixed overhead $(1.60 \times (\$58,500 \div 134,240)) = \0.70

H. Prepare a cost of goods sold budget for the quarter.

Jones Corporation
Cost of Goods Sold Budget

	\$
Direct materials	2517000
Direct labor	805440
Overhead	192740
Add: Beginning inventory	194544
Goods available for sale	3709724
Less: Ending inventory	320544
	\$
Cost of goods sold	3389180

Feedback

Check My Work
Correct
Post-Submission

H.

Direct materials = $(83,900 \times \$30) = \$2,517,000$

Direct labor = $(83,900 \times 1.60 \times \$6) = \$805,440$

Overhead $[(83,900 \times 1.60 \times \$1.00) + \$58,500] = \$192,740$

Beginning inventory $(4,632 \times \$42) = \$194,544$

I. Prepare a budgeted income statement for the quarter—the company falls into the 36 percent tax bracket for income taxes.

Jones Corporation
Budgeted Income Statement

<u>Sales</u> ▼ ✓	\$ ✓ 12944000
<u>Cost of goods sold</u> ▼ ✓	✓ 3389180
Gross margin	✓ \$ 9554820
<u>Less: Variable selling and adm. expenses</u> ▼ ✓	✓ 517760
<u>Less: Fixed selling and administrative expenses</u> ▼ ✓	✓ 123900
<u>Operating income</u> ▼ ✓	✓ \$ 8913160
<u>Less: Income tax expense</u> ▼ ✓	✓ 3208738
Net income	✓ \$ 5704422

Feedback

Check My Work
Correct
Post-Submission

I.

Sales $(\$160 \times 80,900) = \$12,944,000$

Feedback

Check My Work
Correct
Solution

1

A. Prepare a sales budget for July, August, and September, and in total for the quarter.

Jones Corporation
Sales Budget

	July	August	September	Total
Units to be sold	19300	35600	26000	80900

Sales price	\$ 160	\$ 160	\$ 160	\$ 160
Total sales dollars	\$ 3088000	\$ 5696000	\$ 4160000	\$ 12944000

B. Prepare a production budget for July, August, and September, and in total for the quarter.

Jones Corporation
Production Budget

	July	August	September	Total
2 Sales	19300	35600	26000	80900
Add: Desired ending inventory	8544	6240	7632	7632
Total needs	27844	41840	33632	88532
Less: Beginning inventory	4632	8544	6240	4632
Units to be produced	23212	33296	27392	83900

C. Prepare a direct materials purchases budget in pounds and dollars for July, August, and September, and in total for the quarter.

Jones Corporation
Direct Materials Purchases Budget

	July	August	September	Total
Units to be produced	23212	33296	27392	83900
x lbs. per unit	2	2	2	2
3 Pounds needed	46424	66592	54784	167800
Desired ending inventory	17314	14244	25600	25600
Total needs	63738	80836	80384	193400
Less: Beginning inventory	20100	17314	14244	20100
Purchases needed	43638	63522	66140	173300
x \$15/lb.	\$ 15	\$ 15	\$ 15	\$ 15
Total purchase cost	\$ 654570	\$ 952830	\$ 992100	\$ 2599500

D. Prepare a direct labor budget in hours and total cost for July, August, and September, and in total for the quarter.

Jones Corporation
Direct Labor Budget

	July	August	September	Total
4 Units to be produced	23212	33296	27392	83900
x 1.6 of direct labor hours per unit	1.60	1.60	1.60	1.60
Direct labor hours needed	37139	53274	43827	134240
x \$6 per hour	\$ 6	\$ 6	\$ 6	\$ 6
Total direct labor cost	\$ 222834	\$ 319644	\$ 262962	\$ 805440

E. Prepare an overhead budget for July, August, and September, and in total for the quarter.

Jones Corporation
Overhead Budget

	July	August	September	Total
--	------	--------	-----------	-------

Labor hours	37139	53274	43827	134240
x Variable rate per hour	\$ 1.00	\$ 1.00	\$ 1.00	\$ 1.00
Total variable OH	\$ 37139	\$ 53274	\$ 43827	\$ 134240
Fixed overhead	19500	19500	19500	58500
Total overhead	\$ 56639	\$ 72774	\$ 63327	\$ 192740

F. Prepare a selling and administrative expenses budget for July, August, and September, and in total for the quarter.

Jones Corporation
Selling and Administrative Expenses Budget

	July	August	September	Total
Variable selling expense	\$ 123520	\$ 227840	\$ 166400	\$ 517760
Fixed expenses:				
Rent	3600	3600	3600	10800
Utilities	1000	1000	1000	3000
Advertising	200	200	200	600
Office salaries	36500	36500	36500	109500
Total	\$ 164820	\$ 269140	\$ 207700	\$ 641660

G. Prepare an ending finished goods inventory budget for the quarter. [Hint: You have already calculated the desired ending finished goods inventory quantity. Assume a stable per-unit rate. (Round the per-unit fixed factory overhead rate to two decimal places.)]

Jones Corporation
Ending Finished Goods Inventory Budget

Desired ending inventory	\$ 320544
Direct materials	\$ 30
Direct labor	\$ 9.60
Overhead:	
Variable overhead	\$ 1.60
Fixed overhead	\$ 0.70
Unit cost	\$ 42.00

H. Prepare a cost of goods sold budget for the quarter.

Jones Corporation
Cost of Goods Sold Budget

Direct materials	\$ 2517000
Direct labor	805440
Overhead	192740
Add: Beginning inventory	194544
Goods available for sale	3709724
Less: Ending inventory	320544

Cost of goods sold \$ 3389180

I. Prepare a budgeted income statement for the quarter—the company falls into the 36 percent tax bracket for income taxes.

Jones Corporation
Budgeted Income Statement

Sales	\$ 12944000
Cost of goods sold	3389180
Gross margin	\$ 9554820
Less: Variable selling and adm. expenses	517760
Less: Fixed selling and administrative expenses	123900
Operating income	\$ 8913160
Less: Income tax expense	3208738
Net income	\$ <u>5704422</u>

33. PR.09.226.ALGO (Algorithmic)

HairMaster Salons anticipates giving 120 permanents in May, 130 in June, and 140 in July. HairMaster needs one permanent wave kit for each perm, along with two boxes of wave tissues. Its inventory policy is to have 10% of the following month's materials needs on hand. On May 1, there were 18 wave kits and 4 boxes of wave tissues on hand. **(Round any fractions of a unit to the nearest whole unit.)**

A. What is the number of wave kits to be purchased in May?



115
kits

B. What is the number of wave kits to be purchased in June?



131
kits

C. What is the number of boxes of tissues to be purchased in May?



262
boxes

D. What is the number of boxes of tissues to be purchased in June?



262
boxes

Feedback

Check My Work
Correct
Post-Submission

A.
May Wave Kit Purchases
= 120 + 13 - 18= 115

B.

June Wave Kit Purchases

$$= 130 + 14 - 13 = 131$$

C.

May Purchases of Tissue Boxes

$$= (120 \times 2 \text{ boxes}) + (130 \times 2 \text{ boxes} \times 0.10) - 4 = 262$$

D.

June Purchases of Tissue Boxes

$$= (130 \times 2 \text{ boxes}) + (140 \times 2 \text{ boxes} \times 0.10) - (130 \times 2 \text{ boxes} \times 0.10) = 262$$

Solution

HairMaster Salons anticipates giving 120 permanents in May, 130 in June, and 140 in July. HairMaster needs one permanent wave kit for each perm, along with two boxes of wave tissues. Its inventory policy is to have 10% of the following month's materials needs on hand. On May 1, there were 18 wave kits and 4 boxes of wave tissues on hand. **(Round any fractions of a unit to the nearest whole unit.)**

A. What is the number of wave kits to be purchased in May?

kits

B. What is the number of wave kits to be purchased in June?

kits

C. What is the number of boxes of tissues to be purchased in May?

boxes

D. What is the number of boxes of tissues to be purchased in June?

boxes

34. PR.09.227.ALGO (Algorithmic)

Foster Company makes power tools. The sales budget for drills for the first four months of the year is as follows:

Month	Unit Sales
January	29,000
February	18,000
March	26,000
April	26,000

Foster has taken a just-in-time approach to production and wants only 6% of the next month's sales needs in ending inventory. January 1 inventory of drills was zero. Each drill takes 15 minutes of direct labor at \$24 per hour. The factory overhead formula is \$20,000 + \$1.30 per direct labor hour.

Required: Round your answers to the nearest whole unit or whole dollar.

A. What is the budgeted production for January?



30080
drills

B. What is the budgeted production for February?



18480
drills

C. What is the budgeted production for the entire first quarter of the year?



74560
drills

D. What is the budgeted direct labor cost for January?

\$

180480

E. What is the budgeted direct labor cost for February?

\$

110880

F. What is the budgeted variable overhead for March?

\$

8450

G. What is the budgeted total overhead for March?

\$

28450

Feedback

Check My Work
Correct
Post-Submission

A.

Budgeted Production for January
 $29,000 + 1,080 - 0 = 30,080$

B.

Budgeted Production for February
 $18,000 + 1,560 - 1,080 = 18,480$

C.

Budgeted Production for Entire First Quarter of Year
 $30,080 + 18,480 + 26,000^* = 74,560$
 $*26,000 + 1,560 - 1,560$

D.

Budgeted Direct Labor Cost for January
 $30,080 \times 0.25^* \times \$24 = \$180,480$
 $*15 \div 60$

E.

Budgeted Direct Labor Cost for February
 $18,480 \times 0.25 \times \$24 = \$110,880$

F.

Budgeted Variable Overhead for March
 $26,000 \times 0.25 \times \$1.30 = \$8,450$

G.

March Budgeted Total Overhead

$$\$20,000 + \$8,450 = \$28,450$$

Solution

Foster Company makes power tools. The sales budget for drills for the first four months of the year is as follows:

Month Unit Sales

January 29,000

February 18,000

March 26,000

April 26,000

Foster has taken a just-in-time approach to production and wants only 6% of the next month's sales needs in ending inventory. January 1 inventory of drills was zero. Each drill takes 15 minutes of direct labor at \$24 per hour. The factory overhead formula is $\$20,000 + \1.30 per direct labor hour.

Required: Round your answers to the nearest whole unit or whole dollar.

A. What is the budgeted production for January?

30080

drills

B. What is the budgeted production for February?

18480

drills

C. What is the budgeted production for the entire first quarter of the year?

74560

drills

D. What is the budgeted direct labor cost for January?

\$ 180480

E. What is the budgeted direct labor cost for February?

\$ 110880

F. What is the budgeted variable overhead for March?

\$ 8450

G. What is the budgeted total overhead for March?

\$ 28450

35. PR.09.230.ALGO (Algorithmic)

You have decided to throw a party next weekend for 19 friends. The friends are going to bring healthy food, so all you need to do is buy the drinks. You estimate that, on average, each person will drink 3 bottles of soft drinks. Three of your friends will drink only natural soda without unneeded color, so Sulo Ginger Ale should work well for them. For the others and yourself, you decide to buy Sulo Cola. Before going online, you check the refrigerator—you already have 6 bottles of Sulo Ginger Ale and 12 of Sulo Cola. Since this is the end of the semester, you decide that you don't really want any of the soft drinks on hand after the party. Now, you are ordering on the Internet.

A. How many bottles of Sulo Ginger Ale do you plan to buy?



3
bottles

B. How many bottles of Sulo Cola do you plan to buy?



39
bottles

Feedback

Check My Work
Correct
Post-Submission

Ginger Ale Cola

To drink at the party:

3 friends × 3 bottles	9	
17 friends × 3 bottles	51	
+ Desired ending inventory	0	0
Bottles needed	9	51
– Beginning inventory	6	12
Bottles to purchase	3	39

Solution

You have decided to throw a party next weekend for 19 friends. The friends are going to bring healthy food, so all you need to do is buy the drinks. You estimate that, on average, each person will drink 3 bottles of soft drinks. Three of your friends will drink only natural soda without unneeded color, so Sulo Ginger Ale should work well for them. For the others and yourself, you decide to buy Sulo Cola. Before going online, you check the refrigerator—you already have 6 bottles of Sulo Ginger Ale and 12 of Sulo Cola. Since this is the end of the semester, you decide that you don't really want any of the soft drinks on hand after the party. Now, you are ordering on the Internet.

A. How many bottles of Sulo Ginger Ale do you plan to buy?

3

bottles

B. How many bottles of Sulo Cola do you plan to buy?

39

bottles

36. PR.09.232.ALGO (Algorithmic)

Rodelli Company usually sells about 20% of its merchandise during a month for cash with the remaining sales on account. The company's accounts receivable payment history is as follows: 30% in the month of sale, 50% in the month following, and 15% in the second month following the sale. Total budgeted sales for the second quarter are as follows:

April \$101,000
May 144,000
June 74,000

A. What are the expected cash sales in June?

\$

14800

B. What are the expected receipts in June from accounts receivable for sales made in April?

\$

12120

C. What are the expected receipts in June from accounts receivable for sales made in May?

\$

57600

D. What are the total expected cash receipts in June?

\$

102280

E. From the given accounts receivable history information, receipts from accounts receivable do not equal 100%. Does this amount appear on the cash budget?

No

Feedback

Check My Work
Correct
Post-Submission

A.

June Cash Sales = \$74,000 × 0.20 = \$14,800

B.

Receipts on Accounts Receivable for Sales Made in April = 0.80 × \$101,000 × 0.15 = \$12,120

C.

Receipts on Accounts Receivable for Sales Made in May = 0.80 × \$144,000 × 0.50 = \$57,600

D.

Total Cash Expected in June = \$14,800 + \$12,120 + \$57,600 + (0.80 × \$74,000 × 0.30) = \$102,280

E.

30% + 50% + 15% = 95%

The remaining 5% is uncollectible. It does not appear on the cash budget because there is no cash involved.

Solution

Rodelli Company usually sells about 20% of its merchandise during a month for cash with the remaining sales on account. The company's accounts receivable payment history is as follows: 30% in the month of sale, 50% in the month following, and 15% in the second month following the sale. Total budgeted sales for the second quarter are as follows:

April \$101,000

May 144,000

June 74,000

A. What are the expected cash sales in June?

\$

B. What are the expected receipts in June from accounts receivable for sales made in April?

\$

C. What are the expected receipts in June from accounts receivable for sales made in May?

\$

D. What are the total expected cash receipts in June?

\$

E. From the given accounts receivable history information, receipts from accounts receivable do not equal 100%. Does this amount appear on the cash budget?

No

37. PR.09.238.ALGO (Algorithmic)

It is May 28 and you were just hired for a summer job that will pay you (net of taxes) \$810 per month. You start on June 1 and will work until school starts—halfway through August. Your scholarship pays for tuition, room, and board. But you must buy books; pay for transportation to and from school; and pay for clothing, any extra meals, entertainment, and so on. You have gathered the following data:

- 1. One round-trip airline ticket is \$260, and you'd like to come home for Thanksgiving. (Your parents will drive you there in August, and you will try to catch a ride home with another student in December).
- 2. Books are estimated to cost about \$570 per semester for your anticipated major.
- 3. Supplies should be another \$170.
- 4. Clothing might run \$100—you already have almost everything you think you'll need.
- 5. There are 16 weeks in the semester, and you think you'll need \$49 per week for allowance to cover extra meals and entertainment.
- 6. Before school even starts, you need to cover any summer expenses, including going out with friends. \$25 a week sounds about right since all your friends will be working and saving for college as well. There are 11 weeks of summer.

Right now, you have \$230 in your checking account.

Prepare a cash budget for the summer and the first semester of college. (Do the entire time period; do not break it down by week or by month.)

Cash Budget	
Line Item Description	Amount
Beginning balance checking account	\$ 230
Salary from summer job	2025
Cash available	\$ 2255
Less: Disbursements	
Airline ticket	260
Books	570
Supplies	170
Clothing	100
Weekly allowance at school	784
Weekly allowance during the summer	275
Total disbursements	\$ 2159

Estimated ending balance

\$ 96

Feedback

Check My Work
Correct
Post-Submission

Salary from summer job $(2.5 \times \$810) = \$2,025$
Weekly allowance at school $(16 \times \$49) = \784
Weekly allowance during the summer $(11 \times \$25) = \275

Solution

It is May 28 and you were just hired for a summer job that will pay you (net of taxes) \$810 per month. You start on June 1 and will work until school starts—halfway through August. Your scholarship pays for tuition, room, and board. But you must buy books; pay for transportation to and from school; and pay for clothing, any extra meals, entertainment, and so on. You have gathered the following data:

- 1. One round-trip airline ticket is \$260, and you'd like to come home for Thanksgiving. (Your parents will drive you there in August, and you will try to catch a ride home with another student in December).
- 2. Books are estimated to cost about \$570 per semester for your anticipated major.
- 3. Supplies should be another \$170.
- 4. Clothing might run \$100—you already have almost everything you think you'll need.
- 5. There are 16 weeks in the semester, and you think you'll need \$49 per week for allowance to cover extra meals and entertainment.
- 6. Before school even starts, you need to cover any summer expenses, including going out with friends. \$25 a week sounds about right since all your friends will be working and saving for college as well. There are 11 weeks of summer.

Right now, you have \$230 in your checking account.

Prepare a cash budget for the summer and the first semester of college. (Do the entire time period; do not break it down by week or by month.)

Cash Budget

Line Item Description	Amount
Beginning balance checking account	\$ 230
Salary from summer job	2025
Cash available	\$ 2255
Less: Disbursements	
Airline ticket	260
Books	570
Supplies	170
Clothing	100
Weekly allowance at school	784
Weekly allowance during the summer	275
Total disbursements	\$ 2159
Estimated ending balance	\$ 96

Larry Miller, the controller for Kipling Company, has been instructed to develop a flexible budget for overhead costs. The company produces two types of frozen desserts: icy and tasty. The two desserts use common raw materials in different proportions. The company expects to produce 200,000 gallons of each product during the coming year. Icy requires 0.25 direct labor hour per gallon, and Tasty requires 0.30. Larry has developed the following fixed and variable costs for each of the four overhead items:

<u>Overhead Item</u>	<u>Fixed Cost</u>	<u>Variable Rate per DLH</u>
Maintenance	\$52,000	\$1.20
Power		1.50
Indirect labor	79,500	4.80
Rent	54,000	

Required:

- a. Prepare an overhead budget for the expected activity level for the coming year.

Kipling Company Overhead Budget For the Coming Year		
	Formula	✓ 110000 hrs
Variable costs:		
Maintenance	\$1.20	\$ ✓ 132000
Power	1.50	✓ 165000
Indirect labor	4.80	✓ 528000
Total variable costs		✓ \$ 825000
Fixed costs:		
Maintenance		\$ ✓ 52000
Indirect labor		✓ 79500
Rent		✓ 54000
Total fixed costs		\$ ✓ 185500
Total overhead costs		✓ \$ 1010500

Feedback

Check My Work
Correct
Post-Submission

a.

Activity Level*

*Icely: (0.25 × 200,000) 50,000

Tasty: (0.30 × 200,000) 60,000

Total DLH 1,10,000

b. Prepare an overhead budget that reflects production that is 10% higher than expected (for both products). Assume this quantity is within the relevant range.

Kipling Company
Overhead Budget
For the Coming Year

	Formula	✓ 121000 hrs
Variable costs:		
Maintenance	\$1.20	\$ ✓ 145200
Power	1.50	✓ 181500
Indirect labor	4.80	✓ 580800
Total variable costs		\$ ✓ 907500
Fixed costs:		
Maintenance		\$ ✓ 52000
Indirect labor		✓ 79500
Rent		✓ 54000
Total fixed costs		✓ 185500
Total overhead costs		\$ ✓ 1093000

Feedback

Check My Work

Correct

Post-Submission

b.

121,000 Hours*

*110,000 DLH × 110% = 121,000

c. Prepare an overhead budget that reflects production that is 10% lower than expected (for both products). Assume this quantity is within the relevant range.

Kipling Company Overhead Budget For the Coming Year		
	Formula	99000 hrs
Variable costs:		
Maintenance	\$1.20	\$ 118800
Power	1.50	148500
Indirect labor	4.80	475200
Total variable costs		\$ 742500
Fixed costs:		
Maintenance		\$ 52000
Indirect labor		79500
Rent		54000
Total fixed costs		185500
Total overhead costs		\$ 928000

Feedback

Check My Work

Correct

Post-Submission

c.
99,000 Hours*
*110,000 DLH × 90% = 99,000

Feedback

Check My Work

Correct

Solution

1

a. Prepare an overhead budget for the expected activity level for the coming year.

Kipling Company
Overhead Budget
For the Coming Year

	Formula	110000 hrs
Variable costs:		
Maintenance	\$1.20	\$ 132000
Power	1.50	165000
Indirect labor	4.80	528000
Total variable costs		\$ 825000
Fixed costs:		
Maintenance		\$ 52000
Indirect labor		79500
Rent		54000
Total fixed costs		\$ 185500
Total overhead costs		\$ 1010500

b. Prepare an overhead budget that reflects production that is 10% higher than expected (for both products). Assume this quantity is within the relevant range.

Kipling Company
Overhead Budget
For the Coming Year

	Formula	121000 hrs
Variable costs:		
Maintenance	\$1.20	\$ 145200
Power	1.50	181500
Indirect labor	4.80	580800
Total variable costs		\$ 907500
Fixed costs:		
Maintenance		\$ 52000
Indirect labor		79500
Rent		54000
Total fixed costs		185500
Total overhead costs		\$ 1093000

c. Prepare an overhead budget that reflects production that is 10% lower than expected (for both products). Assume this quantity is within the relevant range.

Kipling Company
Overhead Budget
For the Coming Year

	Formula	99000 hrs
Variable costs:		
Maintenance	\$1.20	\$ 118800
Power	1.50	148500

Indirect labor	4.80	475200
Total variable costs		\$ 742500
Fixed costs:		
Maintenance		\$ 52000
Indirect labor		79500
Rent		54000
Total fixed costs		185500
Total overhead costs		\$ 928000